

NSS COLLEGE APPLICATION

ENTITY-RELATIONSHIP (ER) MODEL

B.Tech College — National Service Scheme (NSS)

Editable ER Model

Open the editable ER model in FigJam: https://www.figma.com/online-whiteboard/create-diagram/92435efc-9fb4-4b2b-b272-e66c9d61dfcf?utm_source=chatgpt&utm_content=edit_in_figjam&oi:_id=v1%2Fn2PerVvQNouTXzTJWR2qHHa8j7oN3fyW5kQuJzf9Sfu95AEgbbx4tb&request:_id=7f2cdd2b-5acd-421d-bb76-2ea65e0c3eee

Entities

Entity	Key attributes
USERS	user_id (PK), name, email, phone, status
VOLUNTEERS	volunteer_id (PK), user_id (FK), college_id, department, year, join_date
ROLES	role_id (PK), name
USER_ROLES	user_role_id (PK), user_id (FK), role_id (FK)
NSS_UNITS	unit_id (PK), officer_id (FK), unit_name, unit_number
UNIT_MEMBERSHIPS	membership_id (PK), volunteer_id (FK), unit_id (FK), joined_at, left_at
EVENTS	event_id (PK), unit_id (FK), created_by (FK), title, event_type, start_at, end_at, venue, capacity, status
EVENT_REGISTRATIONS	registration_id (PK), event_id (FK), volunteer_id (FK), registered_at, status
ATTENDANCE_SESSIONS	session_id (PK), event_id (FK), opened_by (FK), starts_at, expires_at, status, token_hash
ATTENDANCE_RECORDS	attendance_id (PK), session_id (FK), volunteer_id (FK), checked_in_at, status
SERVICE_HOUR_ENTRIES	entry_id (PK), volunteer_id (FK), event_id (FK), attendance_id (FK), hours, status, approved_by (FK)
ACTIVITY_REPORTS	report_id (PK), event_id (FK), submitted_by (FK), status, submitted_at
ACHIEVEMENTS	achievement_id (PK), volunteer_id (FK), title, description, status, awarded_at
CERTIFICATES	certificate_id (PK), volunteer_id (FK), certificate_number, status, issued_at
ANNOUNCEMENTS	announcement_id (PK), created_by (FK), unit_id (FK), title, published_at, expires_at
NOTIFICATIONS	notification_id (PK), user_id (FK), title, message, is_read, created_at
DOCUMENTS	document_id (PK), storage_key, file_name, mime_type, size_bytes
AUDIT_LOGS	audit_id (PK), actor_user_id (FK), action, entity_type, entity_id, created_at

Relationships

- USERS 1 — 0..1 VOLUNTEERS: a user may have one volunteer profile.
- USERS 1 — N USER_ROLES and ROLES 1 — N USER_ROLES: users can have assigned roles.
- NSS_UNITS 1 — N UNIT_MEMBERSHIPS and VOLUNTEERS 1 — N UNIT_MEMBERSHIPS: volunteers can have unit membership history.
- NSS_UNITS 1 — N EVENTS: a unit organizes events; authorized users create events.
- EVENTS 1 — N EVENT_REGISTRATIONS and VOLUNTEERS 1 — N EVENT_REGISTRATIONS: volunteers register for events.
- EVENTS 1 — N ATTENDANCE_SESSIONS: an event can have attendance sessions.

- ATTENDANCE_SESSIONS 1 — N ATTENDANCE_RECORDS and VOLUNTEERS 1 — N ATTENDANCE_RECORDS: attendance links volunteers to sessions.
- ATTENDANCE_RECORDS 1 — N SERVICE_HOUR_ENTRIES: verified attendance can generate service-hour entries.
- VOLUNTEERS 1 — N SERVICE_HOUR_ENTRIES: service hours are maintained as a ledger.
- EVENTS 1 — N SERVICE_HOUR_ENTRIES: entries can reference their source event.
- EVENTS 1 — N ACTIVITY_REPORTS: reports can be associated with events.
- VOLUNTEERS 1 — N ACHIEVEMENTS and CERTIFICATES: volunteers can receive multiple records.
- USERS 1 — N NOTIFICATIONS: notifications belong to users.
- ACTIVITY_REPORTS / ACHIEVEMENTS / CERTIFICATES can reference DOCUMENTS for stored artifacts.
- USERS 1 — N AUDIT_LOGS: administrative and security actions are attributed to an actor.

ER Diagram Source

```
erDiagram
    USERS ||--o| VOLUNTEERS : has
    USERS ||--o{ USER_ROLES : assigned
    ROLES ||--o{ USER_ROLES : contains
    NSS_UNITS ||--o{ UNIT_MEMBERSHIPS : has
    VOLUNTEERS ||--o{ UNIT_MEMBERSHIPS : belongs
    NSS_UNITS ||--o{ EVENTS : organizes
    EVENTS ||--o{ EVENT_REGISTRATIONS : has
    VOLUNTEERS ||--o{ EVENT_REGISTRATIONS : registers
    EVENTS ||--o{ ATTENDANCE_SESSIONS : has
    ATTENDANCE_SESSIONS ||--o{ ATTENDANCE_RECORDS : contains
    VOLUNTEERS ||--o{ ATTENDANCE_RECORDS : checks_in
    ATTENDANCE_RECORDS ||--o{ SERVICE_HOUR_ENTRIES : generates
    VOLUNTEERS ||--o{ SERVICE_HOUR_ENTRIES : earns
    EVENTS ||--o{ SERVICE_HOUR_ENTRIES : sources
    EVENTS ||--o{ ACTIVITY_REPORTS : has
    VOLUNTEERS ||--o{ ACHIEVEMENTS : earns
    VOLUNTEERS ||--o{ CERTIFICATES : receives
    USERS ||--o{ NOTIFICATIONS : receives
    USERS ||--o{ AUDIT_LOGS : creates
    ACTIVITY_REPORTS ||--o{ DOCUMENTS : uses
    ACHIEVEMENTS ||--o{ DOCUMENTS : uses
    CERTIFICATES ||--o| DOCUMENTS : references
```

Design Notes

- PostgreSQL is the authoritative store for identity, participation and audit records.
- Service hours are modeled as ledger entries rather than a single mutable total.
- Registration and attendance require uniqueness constraints to prevent duplicates.
- Event capacity must be enforced transactionally to handle concurrent registrations.
- Attendance QR tokens should be short-lived and validated by the backend.
- Audit records should capture critical administrative changes.
- Document metadata remains in the relational database while file contents may be stored in object storage.

Editable Canvas

FigJam: https://www.figma.com/online-whiteboard/create-diagram/92435efc-9fb4-4b2b-b272-e66c9d61dfcf?utm_source=chatgpt&utm_content=edit_in_figjam&oai_id=v1%2Fn2PerVvQNouTXzTJWR2qHHa8j7oN3fyW5kQuJzf9Sfu95AEgbbx4tb&request_id=7f2cdd2b-5acd-421d-bb76-2ea65e0c3eee

The canvas contains the editable visual ER diagram and can be exported from FigJam to supported formats.